

Long Nguyen Khac

Robotics Researcher · Control · Autonomous Systems

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Education

Vietnam National University, Hanoi – International School

Hanoi, Vietnam

B.Eng. in Automation and Informatics

Sep. 2021 – Dec. 2026

- **GPA:** 3.71/4.00; ranked in the top 4% of the cohort.
- Completed a one-year gap period for international research and exchange programs in Indonesia, Taiwan and Singapore.
- **Research focus:** robotics, wireless power transfer, and AI.

Research Experience

Water & Energy Research Lab, School of EEE, Nanyang Technological University

Singapore

Research Fellow

Feb. 2026 – Present

- Project: Advanced 3D Wireless Power Transfer System for Autonomous Underwater Vehicles.
- Investigating relay-coil-enhanced spatial wireless power transfer for AUV charging, with emphasis on power transfer efficiency, coil configuration, and system-level integration.
- Optimizing the mechanical design of AUVs to support underwater operation, wireless charging integration, and long-endurance autonomy.
- Exploring underwater communication between the transmitter and receiver sides of the system for reliable AUV charging coordination.

AISC Lab, AIM-HI Institute, National Chung Cheng University

Chiayi, Taiwan

Research Intern

May 2025 – Jul. 2025

- Project: Smart Sensing Technology for Smart Manufacturing.
- Developed a data-driven approach to estimate dynamic cutting forces in milling using 12 large-scale datasets.
- Combined nonlinear system analysis with sparse identification methods to model and predict cutting force behavior.
- Achieved moderate prediction accuracy with R^2 between 0.5 and 0.6 compared with ground-truth measurements.

AISC Lab, AIM-HI Institute, National Chung Cheng University

Remote

Remote Research Student

Aug. 2025 – Present

- Project: Prediction of cutting force model using AI and data-driven methods.
- Implementing machine learning models, including Random Forest and XGBoost, using 27 large-scale datasets collected through a Taguchi-designed experiment.
- Developing data-driven modeling pipelines for feature engineering, model training, and cutting force prediction in machining processes.

Intelligent Control and Artificial Intelligence Laboratory, VNUIS

Hanoi, Vietnam

Student Researcher

Sep. 2022 – Present

- Project: Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot.
- Designed the mechanical model and physical prototype of a 3-DOF delta robot, and implemented adaptive fuzzy backstepping control for robust trajectory tracking under uncertainties.
- Reduced joint trajectory deviations to less than 0.15 rad, with stability reached in 0.15–0.2 s in simulation studies.
- Project: Detection of Personal Protective Equipment in Construction Sites.
- Built a dataset of 2,632 construction-site images across five classes and trained YOLOv5s with multiple activation functions for PPE detection.

Publications

Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot

International Conference on Advances in Information and Communication Technology, Springer

2024

- **Long Nguyen Khac** et al.
- Proposed an adaptive fuzzy logic control system integrated with backstepping control for a 3-DOF delta robot.
- Validated robust trajectory tracking performance under uncertainties through MATLAB/Simulink simulation.
- DOI: 10.1007/978-3-031-80943-9_57.

YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites

National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability

2023

- **Long Nguyen Khac** et al.
- Developed a deep learning-based PPE detection system using YOLOv5s and a dataset of 2,632 construction-site images.
- Compared six activation functions and achieved the best result with SiLU: Precision 84.8%, Recall 80.2%, mAP@0.5 83.9%, and mAP@0.95 43.6%.

Teaching Experience

Faculty of Engineering and Technology, VNUIS

Hanoi, Vietnam

Teaching Assistant

Sep. 2023 – Dec. 2024

- Course: Modeling and Simulation of Control Systems.
- Assisted 30 students across 7 laboratory sessions on MATLAB/Simulink for robotics, mechatronics, and process control applications.
- Supported the preparation of laboratory manuals, tutorial exercises, and simulation-based learning materials.
- Course: Microprocessors and Microcontrollers.
- Mentored 30 students through 15 laboratory sessions on Embedded C, Arduino, STM32, Keil uVision, Arduino IDE, and STM32CubeIDE.

Hung STEM Robotics Lab

Hanoi, Vietnam

Robotics Teaching Collaborator

Jul. 2022 – Dec. 2022

- Mentored students for the World Robot Olympiad 2022 Vietnam and supported the development of LEGO-based robotics modules.
- Delivered hands-on STEM robotics sessions using LEGO NXT for students at Grammar Newton International Bilingual School, Hanoi.

Working Experience

ETEK Automation Solutions JSC

Hanoi, Vietnam

Engineering Intern

Mar. 2025 – May 2025

- Project: AGV-based Virtual Simulation for Automated Warehouse Logistics – VINFAST Hai Phong.
- Sketched AGV/AMR traffic intersections using AutoCAD and simulated warehouse logistics operations using Visual Components Premium 4.6.
- Monitored AGV/AMR behavior, identified simulation errors, and supported troubleshooting of automated logistics workflows.

International Programs & Exchange Experience

NTU Singapore Global Connect Fellowship

Research Fellowship

- Selected as one of 39 young scholars worldwide for a fully funded research fellowship at Nanyang Technological University.
- Conducted research on advanced 3D wireless power transfer systems for autonomous underwater vehicles.
- Received the Most Popular Poster Award at the final research poster presentation.

Singapore

Feb. 2026 – Mar. 2026

DiscoverNUS, National University of Singapore

Academic Exchange

- Selected as one of 8 excellent VNU students for a fully funded exchange program at NUS.
- Expanded academic exposure to international engineering education, research culture, and interdisciplinary learning.

Singapore

Aug. 2025 – Dec. 2025

Taiwan Experience Education Program, National Chung Cheng University

Research Internship

- Selected as one of 28 global students for a fully funded research internship supported by the Ministry of Education, Taiwan.
- Worked on smart sensing technology and data-driven modeling for smart manufacturing.

Taiwan

May 2025 – Jul. 2025

ASEAN in Today’s World, Kyushu University

International Study Program

- Selected as one of 4 VNU representatives to participate in the ASEAN-focused international study program.

Indonesia

Feb. 2025

Temasek Foundation SCALE

Leadership Exchange Program

- Selected as one of 20 VNU representatives for a fully funded leadership and community action exchange program.

Vietnam & Singapore

Feb. 2024

Honors & Awards

2026	Fully Funded Scholarship, NUS Enterprise Summer Programme in Entrepreneurship, NUS ASEAN SPARK Fellowship	Singapore
2026	Exchange Scholarship, Kyushu University Spring-Semester Exchange Scholarship, Department of Mechanical Engineering	Japan
2026	Most Popular Poster Award, NTU Singapore Global Connect Fellowship Final Research Poster Presentation	Singapore
2026	Research Fellowship, Selected as one of 39 young scholars worldwide for NTU Singapore Global Connect Fellowship	Singapore
2025	Fully Funded Scholarship, DiscoverNUS Scholarship, National University of Singapore	Singapore
2025	Research Internship Scholarship, Taiwan Experience Education Program, Ministry of Education	Taiwan
2025	International Program, ASEAN in Today’s World, Kyushu University	Indonesia
2024	Scholarship, TOSHIBA Scholarship, TOSHIBA International Foundation	Vietnam
2024	Fully Funded Exchange Program, Temasek Foundation Specialist’s Community Action & Leadership Exchange	Vietnam & Singapore
2022	Fighting Spirit Prize, South-East Asia Circuit and System Society Hackathon	Vietnam
2021–2025	Academic Scholarship, Six-semester recipient for excellent academic performance	VNU

Skills

Research	Robotics, autonomous systems, intelligent control, wireless power transfer, smart sensing, AI-enabled engineering systems
Control	Mathematical modeling, trajectory tracking, PID control, adaptive control, backstepping control, fuzzy logic control, MATLAB/Simulink simulation
Robotics	AUV systems, delta robot, AGV/AMR simulation, mechanical design, embedded system integration
Programming	MATLAB, Python, Embedded C/C++, PLC programming
Software	MATLAB/Simulink, SolidWorks, Visual Components, AutoCAD, PyTorch, TIA Portal, STM32CubeIDE, Keil uVision
AI/Data	Machine learning, data-driven modeling, Random Forest, XGBoost, YOLOv5, dataset preparation, model evaluation
Languages	Vietnamese: Native; English: IELTS 6.5, Writing 7.0; Chinese: Beginner

References

Asst. Prof. Yun Yang

Assistant Professor, School of Electrical and Electronic Engineering

- Email: yun.yang@ntu.edu.sg
- Research area: wireless power transfer, power electronics, and energy systems.

Nanyang Technological University

Prof. Her-Terng Yau

Director, AIM-HI Institute

- Email: yauhertern@gmail.com
- Research area: smart manufacturing, nonlinear systems, AI, and data-driven modeling.

National Chung Cheng University

Assoc. Prof. Nguyen Nhu Tung

Senior Lecturer, Faculty of Engineering and Technology

- Email: tungnn@vnu.edu.vn
- Research area: intelligent control, robotics, and automation.

VNU International School

Dr. Le Xuan Hai

Vice Dean, Faculty of Engineering and Technology

- Email: hailx@vnu.edu.vn
- Research area: modeling and simulation, control systems, and mechatronics.

VNU International School